Add vectors by their components

LEARNING TARGET · S2 I can add vectors and resolve their signed components.

SECTION A KEY VOCABULARY

Component	Signed part along an axis; keep axes fixed.
Resultant	Sum: first tail to final tip.
Magnitude	Nonnegative size from perpendicular parts.
Quadrant	Both component signs locate the direction.

SECTION B KEY EQUATIONS + UNITS

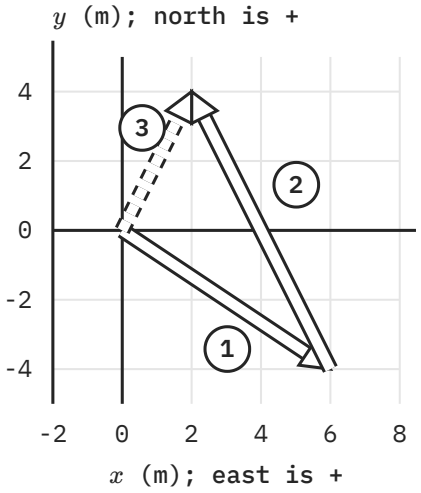
$$R_x = A_x + B_x$$

$$R_y = A_y + B_y$$

$|R|$ $\sqrt{(R_x^2 + R_y^2)}$
angle ratio plus signs

SECTION C COLOR IT · ADD SIGNED PARTS

*Grid steps: **2 m**.*



A = (6, −4) m

B = (−4, 8) m

R = (2, 4) m

$|R|$ = 4.47 m

☐ **1 · BLUE** Displacement A

☐ **2 · BLUE** Displacement B

☐ **3 · BLUE** Resultant R: dashed

SECTION D WORKED EXAMPLE

Two displacements are A = (6, −4) m and B = (−4, 8) m. **Find their sum's magnitude and direction.**

GIVEN	EQUATION	WORK	ANSWER
+x is *east* +y is *north* A and B use these axes.	Add x and y. Use the right triangle.	R = (2, 4) m $\|R\|$ = $\sqrt{\mathbf{20}}$ **m** θ = $\tan^{-1}(4/2)$	**4.47 m** **63.4°** *north* of *east*.

SECTION E THE 2026-XJ THREAD

DAY 6 OF 22 · NEW TOOL: VECTOR SUMS

XJ's motion must be tracked in more than one direction. Adding signed components keeps sideways motion in the forecast instead of treating every displacement as motion toward Earth.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else